# MSG No. 95.135 $P4_322$ [ Type I, tetragonal ]

Table 1: Wyckoff site: `4a`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\, y,\, 0]$ | `[1,5]` |
| 2 | $\left[-y,\, 0,\, \frac{3}{4}\right]$ | `[2,8]` |
| 3 | $\left[y,\, 0,\, \frac{1}{4}\right]$ | `[3,7]` |
| 4 | $\left[0,\, -y,\, \frac{1}{2}\right]$ | `[4,6]` |

Table 2: Wyckoff site: `4b`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\, y,\, 0\right]$ | `[1,5]` |
| 2 | $\left[-y,\, \frac{1}{2},\, \frac{3}{4}\right]$ | `[2,8]` |
| 3 | $\left[y,\, \frac{1}{2},\, \frac{1}{4}\right]$ | `[3,7]` |
| 4 | $\left[\frac{1}{2},\, -y,\, \frac{1}{2}\right]$ | `[4,6]` |

Table 3: Wyckoff site: `4c`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\, x,\, \frac{5}{8}\right]$ | `[1,7]` |
| 2 | $\left[-x,\, x,\, \frac{3}{8}\right]$ | `[2,5]` |
| 3 | $\left[x,\, -x,\, \frac{7}{8}\right]$ | `[3,4]` |
| 4 | $\left[-x,\, -x,\, \frac{1}{8}\right]$ | `[6,8]` |

Table 4: Wyckoff site: `8d`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | `[1]` |
| 2 | $\left[-y,\, x,\, z+\frac{3}{4}\right]$ | `[2]` |
| 3 | $\left[y,\, -x,\, z+\frac{1}{4}\right]$ | `[3]` |
| 4 | $\left[x,\, -y,\, \frac{1}{2}-z\right]$ | `[4]` |
| 5 | $[-x,\, y,\, -z]$ | `[5]` |
| 6 | $\left[-x,\, -y,\, z+\frac{1}{2}\right]$ | `[6]` |
| 7 | $\left[y,\, x,\, \frac{1}{4}-z\right]$ | `[7]` |
| 8 | $\left[-y,\, -x,\, \frac{3}{4}-z\right]$ | `[8]` |